Assignment 4 - Analysis of algorithm SE311T

Total Marks: 100   Questions: 10 x 10 marks   CLOs: CLO3 + CLO4

CLO3: Design new algorithms for different computational problems.
CLO4: Construct proofs of correctness and time complexity of algorithms.

Instructions:

Attempt all questions. Draw DP tables, traversal order, trees, or recurrence clearly. Every answer must include final result and time complexity.

Q1. 0/1 Knapsack DP Table [10 Marks | CLO3, CLO4]

Capacity W = 8. Items: I1(w=2,p=12), I2(w=3,p=20), I3(w=4,p=30), I4(w=5,p=40). Construct the DP table V[i][w], find maximum profit and selected items.

Q2. 0/1 Knapsack Algorithm Design [10 Marks | CLO3, CLO4]

For W = 10 and items A(2,10), B(5,25), C(6,30), D(7,35), design a bottom-up 0/1 knapsack solution. Write recurrence, solve, and give time/space complexity.

Q3. Travelling Salesman Problem [10 Marks | CLO3, CLO4]

For cities A, B, C, D with distances: AB=10, AC=15, AD=20, BC=35, BD=25, CD=30. Find the minimum cost tour starting and ending at A. Also write complexity of brute force.

Q4. TSP Dynamic Programming Recurrence [10 Marks | CLO3, CLO4]

Design the DP recurrence for TSP using subset states. Define the state, base case, transition, final answer, and time complexity.

Q5. Longest Common Subsequence [10 Marks | CLO3, CLO4]

Find the LCS of X = ABCBDAB and Y = BDCABA using dynamic programming. Show the DP length table and one valid LCS.

Assignment 4 - Continued

Q6. LCS Recurrence and Backtracking [10 Marks | CLO3, CLO4]

For X = 10010101 and Y = 01011010, write the LCS recurrence, compute the LCS length, and show one possible LCS by backtracking.

Q7. BFS Traversal [10 Marks | CLO3, CLO4]

Graph edges: AB, AC, BD, BE, CF, EF, DG, FG. Starting from A, apply BFS using alphabetical neighbor order. Give queue steps, parent tree, shortest distances, and complexity.

Q8. DFS Traversal [10 Marks | CLO3, CLO4]

On the same graph in Q7, apply DFS from A using alphabetical neighbor order. Give discovery order, discovery/finish times, DFS tree edges, and complexity.

Q9. Huffman Coding [10 Marks | CLO3, CLO4]

Construct a Huffman tree for A:5, B:9, C:12, D:13, E:16, F:45. Write the code of each character and compute total encoded bits.

Q10. Huffman Coding + Correctness [10 Marks | CLO3, CLO4]

Construct Huffman codes for P:8, Q:12, R:18, S:25, T:37. Compute average code length. Briefly justify why the code is prefix-free and give complexity.